

Photosynthesis and Plant Biology

Photosynthesis is the process by which green plants make their food using sunlight. It takes place in the chloroplasts of plant cells, where chlorophyll absorbs light energy.

Carbon dioxide and water are the raw materials needed for photosynthesis. During the process, plants convert these materials into glucose and oxygen.

Oxygen produced during photosynthesis is released into the atmosphere, which is essential for the survival of most living organisms.

The rate of photosynthesis can be affected by various factors, including light intensity, temperature, and the availability of carbon dioxide.

Leaves are the primary sites of photosynthesis in plants. They are adapted for this function by having a large surface area and containing many chloroplasts.

Stomata are small openings on the underside of leaves that allow for gas exchange. They open during the day to allow carbon dioxide to enter and oxygen to exit.